Topology Optimization
Lecture 12

Local versus global stress constraint strategies in topology optimization: A comparative study

G. A. da Silva, N. Aage, A. T. Beck & O. Sigmund

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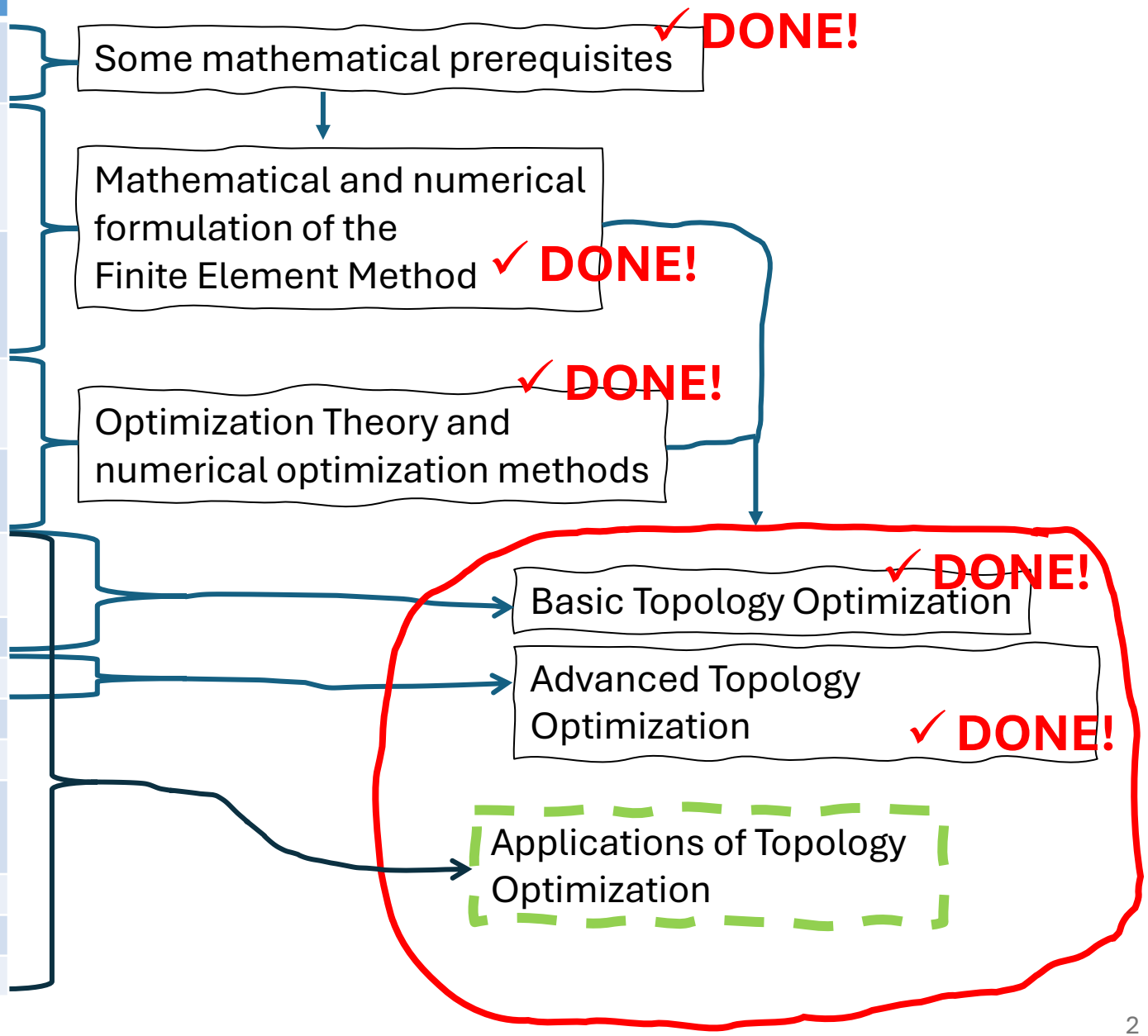
Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction



Some points from last time…

One global constraint

- The stress constraint

$$\sigma_m(\boldsymbol{x}) \leq f_\alpha(\sigma_y)\ \forall \boldsymbol{x} \in \Omega \quad \Rightarrow \quad \max_{\boldsymbol{x}} g(\boldsymbol{x}) - 1 \leq 0 \quad \text{with} \quad g(\boldsymbol{x}) = \frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)}$$

  is approximated by

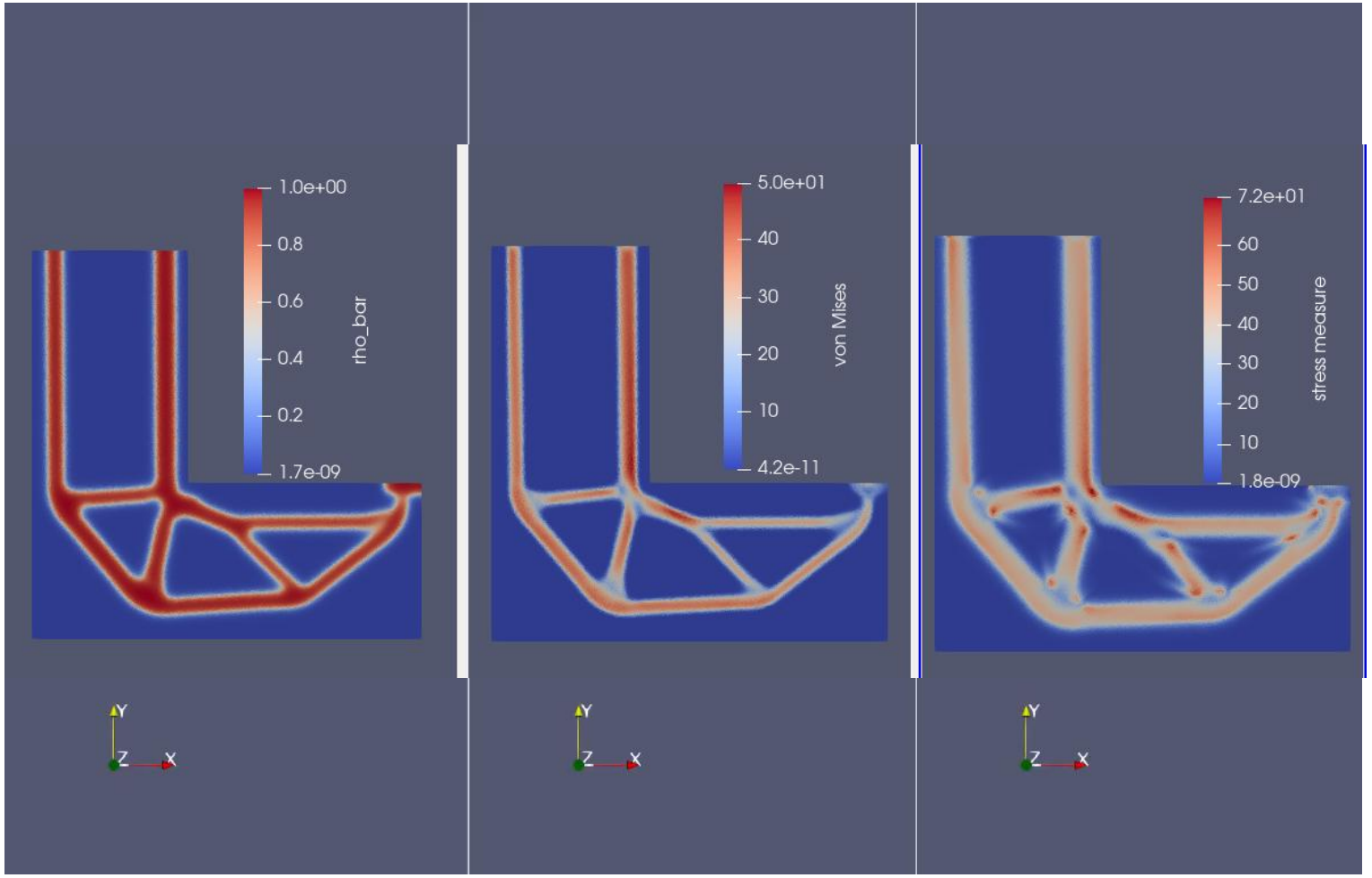
$$\left(\frac{1}{|\Omega|}\int_\Omega \left(\frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)}\right)^p d\Omega\right)^{\frac{1}{p}} - 1 \leq 0 \quad \Rightarrow$$

$$\sigma_c - 1 \leq 0 \quad \text{where} \quad \sigma_c = \left(\frac{1}{|\Omega|}\int_\Omega \left(\frac{\sigma_m(\boldsymbol{x})}{f_\alpha(\sigma_y)}\right)^p d\Omega\right)^{\frac{1}{p}}$$

- The *p*-mean underestimates the stress for low *p* values and the optimization problem is very difficult to solve for high *p*-values.
- It works well in a continuation scheme where *p* is increased gradually, if $\sigma_m(\boldsymbol{x})\ / f_\alpha(\sigma_y)$ does not become much larger than 1.

L-shaped problem

Double filtered robust optimization with global stress constraint, $\sigma_{vM} \leq 70$.



The cost function used here was $\int_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij}d\Omega$ and volume constraint $V_f \leq 0.25$.

β, p, c^b were all updated after 20 MMA iterations.

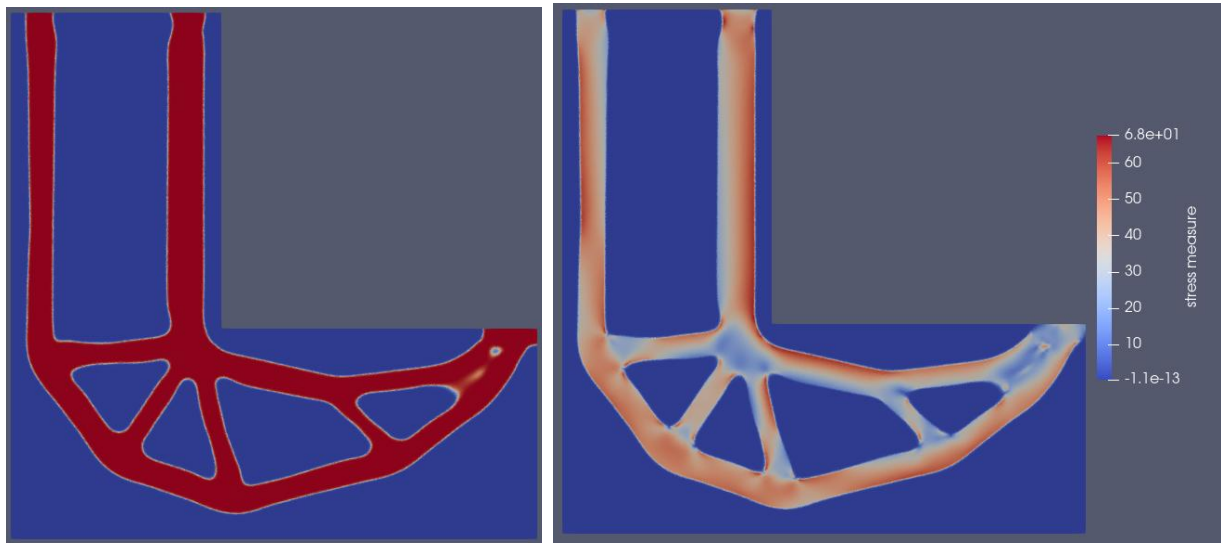
β was raised to $\beta_{\text{lim}}/2$ and p was raised to 300 over 16 step.

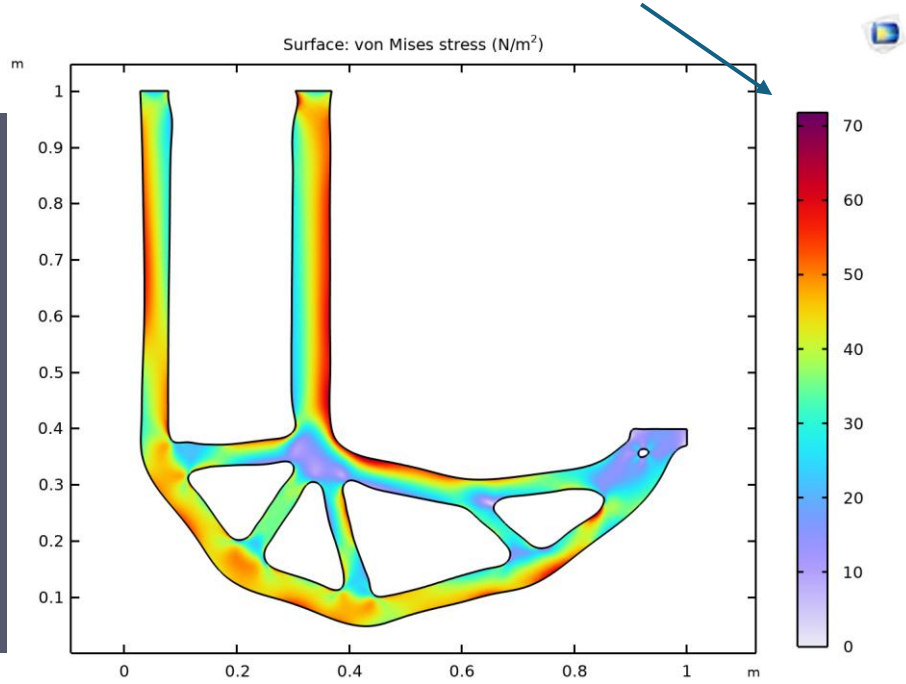
L-shaped example

Maximum only slightly > 70…

Optimized with

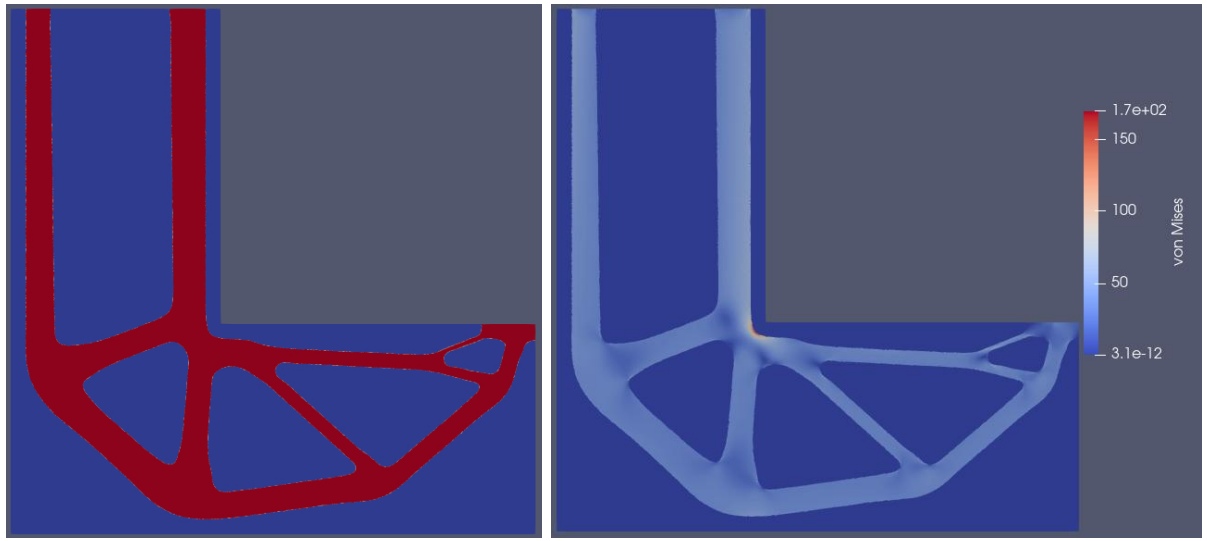
$$\min \int_\Omega \frac{1}{2}\sigma_{ij}^N \epsilon_{ij}^N d\Omega$$
$$\sigma_{vM}^k \leq 70$$
$$k \in \{\mathrm{E}, \mathrm{N}, \mathrm{D}\}$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 14$$
$$p_{\max} = 300$$

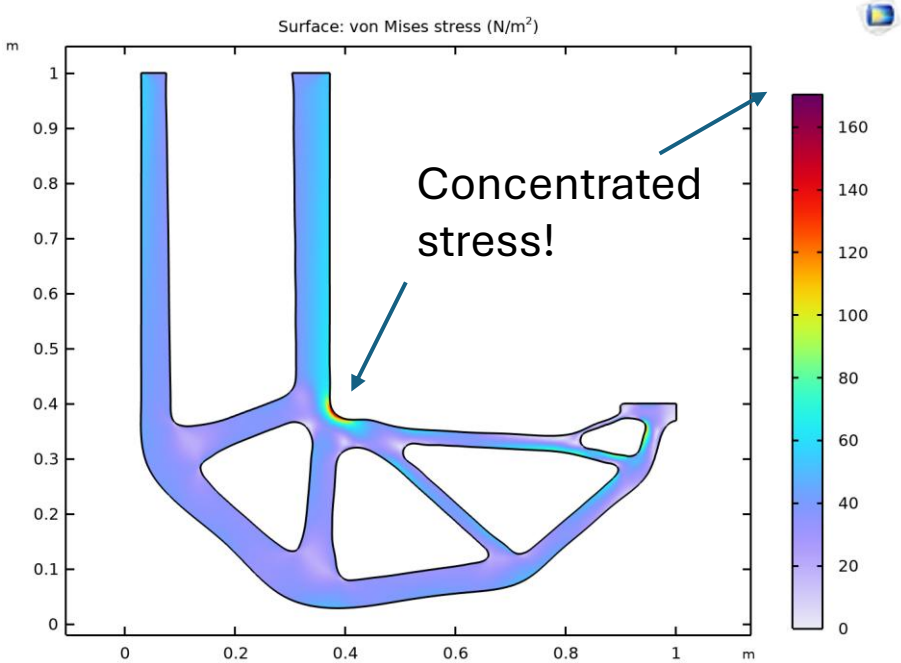




Optimized with

$$\min \max_{k \in \{\mathrm{E},\mathrm{N},\mathrm{D}\}} f^k$$
$$f^k = \int_\Omega \frac{1}{2}\sigma_{ij}^k \epsilon_{ij}^k d\Omega$$
$$V_f \leq 0.3$$
$$\Delta\eta = 0.05$$
$$\beta_{\max} = 64$$





Many local constraints

- Consider an optimization problem with an infinite number of stress constraints

$$\min_{\rho} \mathrm{f}(\rho)\ s.t.\ \frac{\sigma_m\big(\rho(\boldsymbol{x})\big)}{f_\alpha\big(\sigma_y\big)} - 1 \leq 0\ \forall \boldsymbol{x} \in \Omega$$

  and write down the Lagrange function

$$L = f(\rho) + \int_\Omega \lambda(\boldsymbol{x}) \left(\frac{\sigma_m\big(\rho(\boldsymbol{x})\big)}{f_\alpha\big(\sigma_y\big)} - 1 \right) d\Omega$$

  where $\lambda(\boldsymbol{x})$ is a Lagrange multiplier field.

- There is not really any fundamental mathematical issues with an infinite number of constraints (or large finite number of constraints). It is only a problem for constrained optimization algorithms, as the constraints are considered explicitly in the solution process.
- Algorithms which aggregate all the constraints in a single function are able to handle the many constraints.
- Penalty methods and augmented Lagrangian methods do exactly that.

Penalty Methods

To solve a constrained optimization problem using an unconstrained optimization algorithm, one can replace the original cost function by a new cost function which includes a penalty for violated constraints.

Consider the problem with equality constraints only:

$$\min f(\boldsymbol{x}) \text{ subject to } h_i(\boldsymbol{x}) = 0 \quad , \quad i = 1, ..., p$$

The new cost function is defined as:

$$Q(\boldsymbol{x}, \mu) = f(\boldsymbol{x}) + \frac{\mu}{2} \sum_{i=1}^{p} h_i^2(\boldsymbol{x})$$

This is termed the quardratic penalty function for the problem.

Now we do a series of unconstrained minimizations of Q for $\mu = \mu^1, \mu^2, ..., \mu^k$ where $\mu^{i+1} > \mu^i$ for $i = 1, ..., k-1$.

The solution to the $(i-1)$'th problem is used as the starting point for the i'th problem.

As $\mu^k \to \infty$, constraint violations give a larger and larger penalty to the new cost function, so to reduce these penalties the solution should show $h_i \to 0$ as $\mu^k \to \infty$.

Penalty Methods

The penalty function can also be extended to handle inequality constraints.

A popular function for both equality and inequality constraints is:

$$Q(\boldsymbol{x}, \mu) = f(\boldsymbol{x}) + \frac{\mu}{2}\left(\sum_{i=1}^{p} h_i^2(\boldsymbol{x}) + \sum_{i=1}^{m} \left(\max\left(0, g_i(\boldsymbol{x})\right)\right)^2\right)$$

Note that Q above is generally not twice continuous differentiable, as we usually consider functions to be in this course!

Alternatively one could use the quardratic penalty function for equality constraint after having used slack variables to turn the inequality constraints into equality constraints.

Note: Many other penalty (and "barrier") functions are used in practice.

Augmentet Lagrangian Method or the method of multipliers

The problems with the penalty method come from the ill-conditioning as μ become large, which is needed to get a highly accurate solution.

In the penalty method, the constraints are not satisfied exactly at optimum, since from Nocedal&Wright theorem 17.2:

$$h_i(\boldsymbol{x}^k) \approx -\frac{\lambda_i^*}{\mu^k}$$

The augmented Lagrangian method tries to remove the deficiencies of the penalty method by obtaining an accurate solution for a not necessarily very large value of μ.

The augmented Lagrange function is defined as:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \mu^k) = f(\boldsymbol{x}) - \sum_{i=1}^{p} v_i^k h_i(\boldsymbol{x}) + \frac{\mu^k}{2} \sum_{i=1}^{p} h_i^2(\boldsymbol{x})$$

This is a combination of the quardratic penalty function and the usual Lagrange function.

The method fixes both the Lagrange multipliers and the penalty parameter in each iteration to $\boldsymbol{v}^k$ and μ^k.

Then the problem for fixed $\boldsymbol{v}^k$ and μ^k can be solved by an unconstrained optimization method and $\boldsymbol{v}^k$ and μ^k can be updated for the next iteration.

Augmentet Lagrangian Method or the method of multipliers

$w_i^* = v_i^k - \mu^k h_i(\boldsymbol{x}^k)$ suggest the following method for updating the Lagrange multipliers:

$$v_i^{k+1} = v_i^k - \mu^k h_i(\boldsymbol{x}^k)$$

Augmented Lagrangian Method:

Choose some starting parameters, μ^0, $\boldsymbol{x}^{-1}$, and $\boldsymbol{v}^0$.
Choose a starting tolerance, ϵ^0, and stopping criteria, τ and μ_{max}.
Set $k = 0$.
While $\mu^k < \mu_{max}$ and ($k = 0$ or $||\boldsymbol{x}^{k-1} - \boldsymbol{x}^{k-2}|| > \tau$)
    Find $\boldsymbol{x}^k$ by
      minimizing $\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \mu^k)$ wrt. $\boldsymbol{x}$ from starting point $\boldsymbol{x}^{k-1}$ to tolerence ϵ^k.
    $v_i^{k+1} = v_i^k - \mu^k h_i(\boldsymbol{x}^k)$.
    Choose $\mu^{k+1} > \mu^k$ and $\epsilon^{k+1} \leq \epsilon^k$.
    $k = k + 1$.

Often $\mu^{k+1} = K_\mu \mu^k$ and $\epsilon^{k+1} = \epsilon^k / K_\epsilon$ is used for some constants $K_\mu > 1$ and $K_\epsilon \geq 1$.

We often get away with a (much) smaller K_μ than the one used in the quadratic penalty method, but an initial too small K_μ may lead to an unbounded problem.

From Lecture 4

THEOREM 4.6
Karush-Kuhn-Tucker Optimality Conditions

Let $\mathbf{x}^*$ be a regular point of the feasible set that is a local minimum for $f(\mathbf{x})$, subject to $h_i(\mathbf{x}) = 0; i = 1$ to $p; g_j(\mathbf{x}) \le 0; j = 1$ to m. Then there exist Lagrange multipliers $\mathbf{v}^*$ (a p-vector) and $\mathbf{u}^*$ (an m-vector) such that the Lagrangian function is stationary with respect to x_i, v_i, u_j, and s_j at the point $\mathbf{x}^*$.

1. Lagrangian function for the problem written in the standard form:

$$L(\mathbf{x},\mathbf{v},\mathbf{u},\mathbf{s}) = f(\mathbf{x}) + \sum_{i=1}^{p} v_i h_i(\mathbf{x}) + \sum_{j=1}^{m} u_j (g_j(\mathbf{x}) + s_j^2) = f(\mathbf{x}) + \mathbf{v}^{\mathrm{T}} \mathbf{h}(\mathbf{x}) + \mathbf{u}^{\mathrm{T}} (\mathbf{g}(\mathbf{x}) + \mathbf{s}^2) \tag{4.46}$$

2. Gradient conditions:

$$\frac{\partial L}{\partial x_k} = \frac{\partial f}{\partial x_k} + \sum_{i=1}^{p} v_i^* \frac{\partial h_i}{\partial x_k} + \sum_{j=1}^{m} u_j^* \frac{\partial g_j}{\partial x_k} = 0; \quad k = 1 \text{ to } n \tag{4.47}$$

$$\frac{\partial L}{\partial v_i} = 0 \Rightarrow h_i(\mathbf{x}^*) = 0; \quad i = 1 \text{ to } p \tag{4.48}$$

$$\frac{\partial L}{\partial u_j} = 0 \Rightarrow (g_j(\mathbf{x}^*) + s_j^2) = 0; \quad j = 1 \text{ to } m \tag{4.49}$$

3. Feasibility check for inequalities:

$$s_j^2 \ge 0; \text{ or equivalently } g_j \le 0; \quad j = 1 \text{ to } m \tag{4.50}$$

4. Switching conditions:

$$\frac{\partial L}{\partial s_j} = 0 \Rightarrow 2u_j^* s_j = 0; \quad j = 1 \text{ to } m \tag{4.51}$$

5. Nonnegativity of Lagrange multipliers for inequalities:

$$u_j^* \ge 0; \quad j = 1 \text{ to } m \tag{4.52}$$

6. *Regularity check*: Gradients of the active constraints must be linearly independent. In such a case the Lagrange multipliers for the constraints are unique.

TABLE 5.1 Alternate form of KKT necessary conditions

Problem: Minimize $f(\mathbf{x})$ subject to $h_i(\mathbf{x}) = 0$, $i = 1$ to p; $g_j(\mathbf{x}) \le 0$, $j = 1$ to m

1. Lagrangian function definition	$L = f + \sum_{i=1}^{p} v_i h_i + \sum_{j=1}^{m} u_j g_j$	(5.1)
2. Gradient conditions	$\frac{\partial L}{\partial x_k} = 0; \quad \frac{\partial f}{\partial x_k} + \sum_{i=1}^{p} v_i^* \frac{\partial h_i}{\partial x_k} + \sum_{j=1}^{m} u_j^* \frac{\partial g_j}{\partial x_k} = 0; \quad k = 1$ to n	(5.2)
3. Feasibility check	$h_i(\mathbf{x}^*) = 0; \quad i = 1$ to $p; \quad g_j(\mathbf{x}^*) \le 0; \quad j = 1$ to m	(5.3)
4. Switching conditions	$u_j^* g_j(\mathbf{x}^*) = 0; \quad j = 1$ to m	(5.4)
5. Non-negativity of Lagrange multipliers for inequalities	$u_j^* \ge 0; \quad j = 1$ to m	(5.5)
6. Regularity check: Gradients of active constraints must be linearly independent. In such a case, the Lagrange multipliers for the constraints are unique.		

Introduction to Optimum Design, 3rd edition, J. S. Arora, ISBN: 978-0-12-381375-6

Let’s put this to use for handling many stress constraints…

Augmentet Lagrangian Method with inequality constraints

For equality constraints, $h_i = 0$, the augmented Lagrange function is defined as:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \mu^k) = f(\boldsymbol{x}) - \sum_{i=1}^{p} v_i^k h_i(\boldsymbol{x}) + \frac{\mu^k}{2} \sum_{i=1}^{p} h_i^2(\boldsymbol{x}) \qquad = f(\boldsymbol{x}) + \frac{\mu^k}{2} \sum_{i=1}^{p} \left(\left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k} \right)^2 - \left(\frac{v_i^k}{\mu^k} \right)^2 \right)$$

Inequality constraints can be included by using slack variables

$$g_j(\boldsymbol{x}) \leq 0 \;\Rightarrow\; g_j(\boldsymbol{x}) + s_j = 0 \quad , \quad s_j \geq 0$$

For a problem with p equality and q inequality constraints, the augmented Lagrangian become

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k, \mu^k, \boldsymbol{s}^k) = f(\boldsymbol{x}) + \frac{\mu^k}{2} \sum_{i=1}^{p} \left(\left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k} \right)^2 - \left(\frac{v_i^k}{\mu^k} \right)^2 \right) + \frac{\mu^k}{2} \sum_{j=1}^{q} \left(\left(g_j(\boldsymbol{x}) + s_j - \frac{w_j^k}{\mu^k} \right)^2 - \left(\frac{w_j^k}{\mu^k} \right)^2 \right) \quad , \quad s_j \geq 0$$

We now want to eleminate the slack variables. From the KKT conditions, the necessary conditions for an optimum contain

$$\frac{\partial \mathcal{L}}{\partial s_j} = 0 \quad , \quad s_j \geq 0 \Rightarrow \quad \frac{\mu^k}{2} 2 \left(g_j(\boldsymbol{x}) + s_j - \frac{w_i^k}{\mu^k} \right) = 0 \quad , \quad s_j \geq 0 \Rightarrow \quad s_j = \frac{w_j^k}{\mu^k} - g_j(\boldsymbol{x}) \quad , \quad s_j \geq 0$$

This can also be written simply as

$$s_j = \max\left(0, \frac{w_j^k}{\mu^k} - g_j(\boldsymbol{x}) \right) \equiv \left\langle \frac{w_j^k}{\mu^k} - g_j(\boldsymbol{x}) \right\rangle \quad , \quad \langle x \rangle \equiv \max(0, x)$$

Augmentet Lagrangian Method with inequality constraints

Inserting

$$s_j = \max\left(0, \frac{w_j^k}{\mu^k} - g_j(\boldsymbol{x})\right) \equiv \left\langle \frac{w_j^k}{\mu^k} - g_j(\boldsymbol{x}) \right\rangle$$

into

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k, \mu^k, \boldsymbol{s}^k) = f(\boldsymbol{x}) + \frac{\mu^k}{2}\sum_{i=1}^{p}\left(\left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k}\right)^2 - \left(\frac{v_i^k}{\mu^k}\right)^2\right) + \frac{\mu^k}{2}\sum_{j=1}^{q}\left(\left(g_j(\boldsymbol{x}) + s_j - \frac{w_j^k}{\mu^k}\right)^2 - \left(\frac{w_j^k}{\mu^k}\right)^2\right) \quad , \quad s_j \geq 0$$

yield

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k, \mu^k) = f(\boldsymbol{x}) + \frac{\mu^k}{2}\sum_{i=1}^{p}\left(\left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k}\right)^2 - \left(\frac{v_i^k}{\mu^k}\right)^2\right) + \frac{\mu^k}{2}\sum_{j=1}^{q}\left(\left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k}\right\rangle^2 - \left(\frac{w_j^k}{\mu^k}\right)^2\right)$$

since if

$$s_j = 0 \text{ then } \frac{w_j^k}{\mu^k} - g_j(\boldsymbol{x}) \leq 0 \;\Rightarrow g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \geq 0 \text{ and } \left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k}\right\rangle = g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k}$$

$$s_j > 0 \text{ then } s_j = \frac{w_j^k}{\mu^k} - g_j(\boldsymbol{x}) \;\Rightarrow g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} < 0 \text{ and } \left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k}\right\rangle = 0 = \left(g_j(\boldsymbol{x}) + s_j - \frac{w_j^k}{\mu^k}\right)$$

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k, \mu^k) = f(\boldsymbol{x}) + \frac{\mu^k}{2} \sum_{i=1}^{p} \left(\left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k} \right)^2 - \left(\frac{v_i^k}{\mu^k} \right)^2 \right) + \frac{\mu^k}{2} \sum_{j=1}^{q} \left(\left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle^2 - \left(\frac{w_j^k}{\mu^k} \right)^2 \right)$$

Using

$$\left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle^2 = \begin{cases} 0 & \text{if } g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \leq 0 \\ \left(g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right)^2 & \text{if } g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \geq 0 \end{cases}$$

$$\nabla \left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle^2 = \begin{cases} 0 & \text{if } g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \leq 0 \\ 2 \left(g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right) \nabla g_j(\boldsymbol{x}) & \text{if } g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \geq 0 \end{cases} = \max \left(0, g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right) \nabla g_j(\boldsymbol{x}) = \left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle \nabla g_j(\boldsymbol{x})$$

the gradient (wrt. $\boldsymbol{x}$) is

$$\begin{aligned} \nabla \mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k, \mu^k) &= \nabla f(\boldsymbol{x}) + \mu^k \sum_{i=1}^{p} \left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k} \right) \nabla h(\boldsymbol{x}) + \mu^k \sum_{j=1}^{q} \nabla \left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle^2 \\ &= \nabla \mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k, \mu^k) = \nabla f(\boldsymbol{x}) + \mu^k \sum_{i=1}^{p} \left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k} \right) \nabla h(\boldsymbol{x}) + \mu^k \sum_{j=1}^{q} \left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle \nabla g_j(\boldsymbol{x}) \end{aligned}$$

Comparing to the gradient of the Lagrange function

$$L(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k) = f(\boldsymbol{x}) - \sum_{i=1}^{p} v_i^* h_i(\boldsymbol{x}) - \sum_{j=1}^{q} w_j^* g_j(\boldsymbol{x}) \quad , \quad \nabla L(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k) = \nabla f(\boldsymbol{x}) - \sum_{i=1}^{p} v_i^* \nabla h_i(\boldsymbol{x}) - \sum_{j=1}^{q} w_j^* \nabla g_j(\boldsymbol{x})$$

suggest that the Lagrange multipliers satisfies

$$v_i^* = -\mu^k \left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k} \right) = -\left(\mu^k h_i(\boldsymbol{x}) - v_i^k \right) \quad , \quad w_j^* = -\mu^k \left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle = -\left\langle \mu^k g_j(\boldsymbol{x}) - w_j^k \right\rangle$$

Augmentet Lagrangian Method with inequality constraints

Since the augmented Lagrangian is, in each iteration of the algorithm, minimized wrt. $\boldsymbol{x}$ for fixed $\boldsymbol{v}^k$, $\boldsymbol{w}^k$, and μ^k, some terms can be dropped to give

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{v}^k, \boldsymbol{w}^k, \mu^k) = f(\boldsymbol{x}) + \frac{\mu^k}{2} \sum_{i=1}^{p} \left(\left(h_i(\boldsymbol{x}) - \frac{v_i^k}{\mu^k} \right)^2 \right) + \frac{\mu^k}{2} \sum_{j=1}^{q} \left(\left\langle g_j(\boldsymbol{x}) - \frac{w_j^k}{\mu^k} \right\rangle^2 \right)$$

and the multipliers should be updated as $\boldsymbol{v}^k$, $\boldsymbol{w}^k$

$$v_i^{k+1} = -\left(\mu^k h_i(\boldsymbol{x}) - v_i^k\right) \quad , \quad w_j^{k+1} = -\left\langle \mu^k g_j(\boldsymbol{x}) - w_j^k \right\rangle$$

The paper by da Silva only considers inequality constraints and have chosen the opposite sign of the multipliers and uses different names for the variables ($\mu^k \to r^k,\ w_j^k \to -\mu_j^k, \boldsymbol{x} \to \boldsymbol{\rho}$), yielding

$$\mathcal{L}(\boldsymbol{\rho}, \boldsymbol{\mu}^k, r^k) = f(\boldsymbol{\rho}) + \frac{r^k}{2} \sum_{j=1}^{q} \left(\left\langle \frac{\mu_j^k}{r^k} + g_j(\boldsymbol{\rho}) \right\rangle^2 \right) \quad , \quad \mu_j^{k+1} = \left\langle r^k g_j(\boldsymbol{\rho}) + \mu_j^k \right\rangle$$

The algorithm is then simply to solve the following problem in a continuation scheme ($k = 1, 2, ...$)

$$\min_{\boldsymbol{\rho}} \mathcal{L}(\boldsymbol{\rho}, \boldsymbol{\mu}^k, r^k) \quad \text{s.t.}$$

equlibrium equations are satisfied,

filtering and projection is applied etc.

$0 \leq \rho_i \leq 1$

After each (approximate) solution of the above problem, the penalty parameter, r^k and Lagrange multipliers, μ_j^k, are updated and β can be updated as well.

The penalty method is obtained by setting the (approximate) Lagrange multipliers $\mu_j^k = 0$ in all iterations.

Augmentet Lagrangian and penalty methods

- The augmented Lagrangian and penalty methods solves an unconstrained minimization in each iteration.
- In topology optimization, bound constraints are needed.
- Many unconstrained optimization methods (e.g., steepest descent or conjugate gradients) can be extended to handle bound constraints.
- The paper suggest an implementation of steepest descent to minimize the augmented Lagrangian or penalty function.
- The method is named Steepest Descent Method with Move Limits.
- MMA can also be used, just without any constraints.
  - The implementation we use assumes that at least 1 constraint is present.
  - This can be easily be circumvented by specifying 1 “dummy” constraint such as

$$g(\boldsymbol{\rho}) - 1 \leq 0 \text{ with } g(\boldsymbol{\rho}) = 0$$
$$(\text{Note: } \nabla g = \mathbf{0} \text{ (zero vector)})$$

Results

$$\underset{\rho}{\text{Min.}}\ V_f\left(\overline{\boldsymbol{\rho}}\right) = \frac{\sum_{e=1}^{N_e} V_e \overline{\rho}_e}{\sum_{e=1}^{N_e} V_e}$$

$$\text{s.t.}\ \frac{\sigma_{eq}^{(k)}\left(\overline{\boldsymbol{\rho}}\right)}{f_\alpha\left(\sigma_y\right)} - 1 \leqslant 0 \qquad k = 1, 2, \ldots, N_k$$

$$\mathbf{K}\left(\overline{\boldsymbol{\rho}}\right)\mathbf{U}\left(\overline{\boldsymbol{\rho}}\right) = \mathbf{F}$$

$$0 \leqslant \rho_e \leqslant 1 \qquad e = 1, 2, \ldots, N_e,$$

- The paper looks at the following measures:
  - Volume used (the cost function)

$$V_f = \frac{\sum_{e=1}^{N_e} V_e \bar{\rho}_e}{\sum_{e=1}^{N_e} V_e} \sim \frac{1}{|\Omega|}\int_\Omega \bar{\rho}\, d\Omega$$

  - Stress

$$\sigma_{max} = \max \sigma_m(\boldsymbol{x}), \qquad \sigma_m(\boldsymbol{x}) = f_\sigma(\bar{\rho})\sqrt{\sigma_{vM}^2 + \sigma_{\min}^2} = \sigma^{eq},$$

$$f_\sigma(\bar{\rho}) = \frac{\bar{\rho}}{\epsilon(1-\bar{\rho}) + \bar{\rho}}, \qquad \sigma_{vM}^2 = \frac{3}{2} s_{ij} s_{ij}, \qquad s_{ij} = \sigma_{ij} - \frac{1}{3}\sigma_{kk}\delta_{ij}$$

  - "Measure of non-discreteness"

$$M_{nd} = \frac{\sum_{e=1}^{N_e} 4\bar{\rho}_e(1-\bar{\rho}_e)}{N_e} \sim \frac{1}{|\Omega|}\int_\Omega 4\bar{\rho}(1-\bar{\rho})\, d\Omega$$

- The two different algorithms ('global' P-mean and 'local' augmented Lagrangian/penalty) are compared.
  - Many different choices are made in the continuation schemes, algorithms etc.
  - 2000 iterations is done in 2 phases:
    - 1000 iterations in phase 1, the 'parameter increasing phase' where (β, P, r) are changed.
    - followed by 1000 iterations in phase 2, the 'stabilization phase'.
    - Lagrange multipliers μ_i and adaptive constraint normalization c^b is still updated in phase 2 .
  - Late in the paper, a smaller number of iterations in phase 1 is investigated and phase 2 is ended when the calculation is 'converged'.

Local strategies

TABLE 1 Volume fractions and maximum stress constraint violations after 2000 iterations, for all local strategies and both mesh resolutions

		Mesh: 200 × 200		Mesh: 800 × 800	
Solution strategy	r_{max}	V_f (%)	$\sigma_{max} - 1$ (%)	V_f (%)	$\sigma_{max} - 1$ (%)
Local 1: AL + SDM	10^3	23.26	1.31	23.34	15.26
	10^4	**23.74**	**0.15**	**23.95**	**0.39**
	10^5	25.29	0.60	24.68	1.49
	10^6	28.74	0.32	28.01	1.85
Local 2: AL + MMA	10^3	23.27	1.75	22.80	16.78
	10^4	**23.53**	**0.37**	**23.34**	**0.34**
	10^5	25.43	0.44	24.32	0.61
	10^6	31.17	0.72	29.27	1.52
Local 3: EP + SDM	10^3	22.33	22.95	22.26	39.18
	10^4	23.53	2.24	23.48	13.05
	10^5	**24.70**	**0.47**	**24.61**	**0.79**
	10^6	27.55	1.19	28.53	0.64
Local 4: EP + MMA	10^3	22.44	21.26	22.14	38.61
	10^4	23.63	2.10	23.35	13.20
	10^5	**25.02**	**0.60**	**24.45**	**1.36**
	10^6	28.05	0.93	29.51	0.95

Note: Most appropriate r_{max} parameters are highlighted in bold for each strategy.

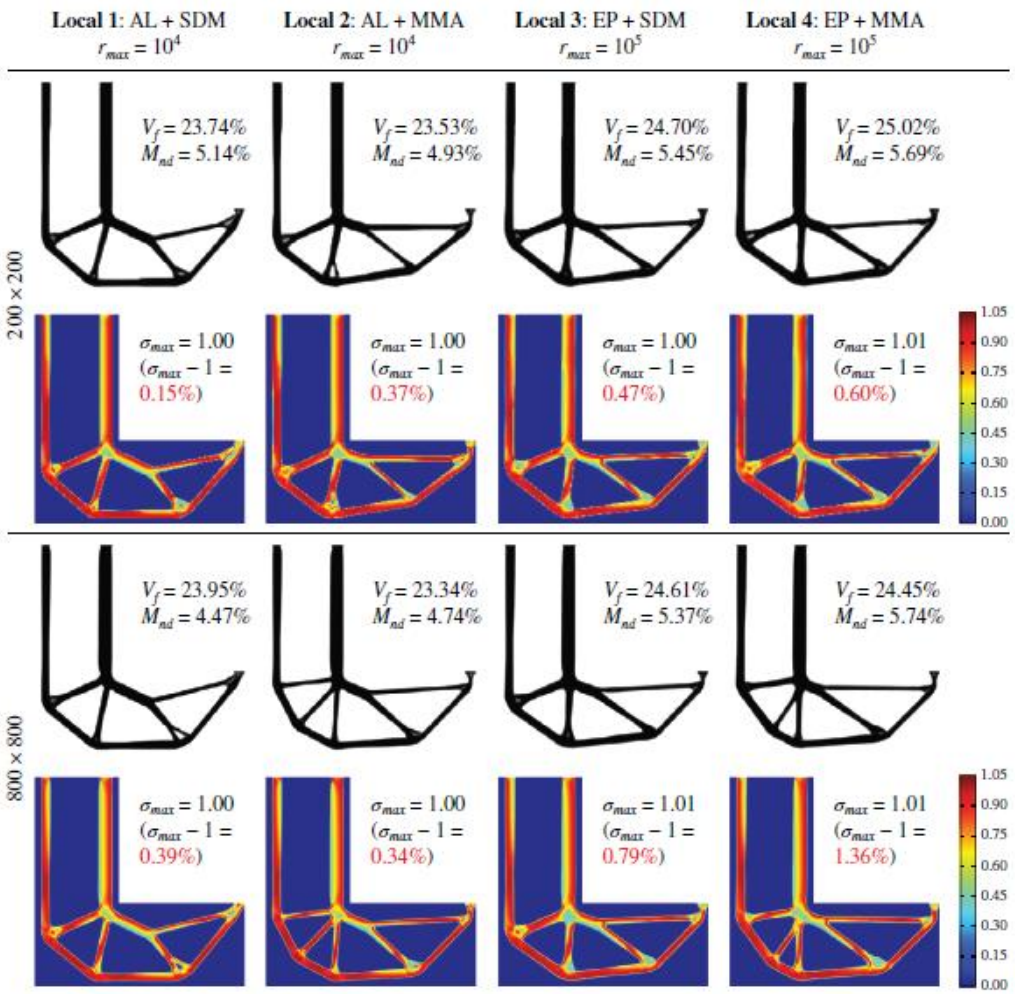


FIGURE 2 Optimized topologies and respective von Mises stresses obtained with the four local strategies for the most promising values of r_{max} and two mesh resolutions, at 2000 iterations

- For the penalty method (EP), higher values of the maximum penalty factor, r_{max}, leads to smaller constraint violations. But too large r_{max} leads to convergence issues, as expected.
- For the augmented Lagrangian (AL) the constraint violation is not only decreasing when the maximum penalty factor r_{max} is increasing. $r_{max} \sim 10^4$ seem to work well for AL.
- A finer mesh does not necessarily lead to the same structure - the final structure may end up being a different local optimum.

Global Strategies

TABLE 2 Volume fractions and maximum stress constraint violations after 2000 iterations, for all global strategies and both mesh resolutions

Solution strategy	P_{max}	Mesh: 200 × 200 V_f (%)	Mesh: 200 × 200 $\sigma_{max} - 1$ (%)	Mesh: 800 × 800 V_f (%)	Mesh: 800 × 800 $\sigma_{max} - 1$ (%)
Global 1: P-continuation	50	20.72	14.74	20.68	18.86
	100	21.68	6.84	21.80	9.37
	300	**22.97**	**2.52**	**23.12**	**2.88**
	500	23.89	2.25	24.39	2.12
Global 2: P-continuation + c_{PM} (1 it)	50	24.35	0.00	26.15	0.07
	100	**24.23**	**0.14**	**25.03**	**0.09**
	200	25.80	0.59	24.90	−0.32
	300	24.40	1.43	27.02	2.75
Global 3: P-continuation + c_{PM} (20 it)	50	24.65	0.01	26.24	0.01
	100	**23.86**	**0.05**	**25.00**	**0.04**
	200	20.99	154.65	24.46	0.08
	300	24.66	0.19	25.53	0.53
Global 4: Fixed P + c_{PM} (1 it)	50	26.13	−0.01	26.42	0.11
	60	**25.44**	**0.16**	**25.85**	**−0.02**
	80	27.38	0.12	27.01	0.09
	100	27.55	−0.03	28.04	0.04

Note: Most appropriate P_{max} parameters are highlighted in bold for each strategy.

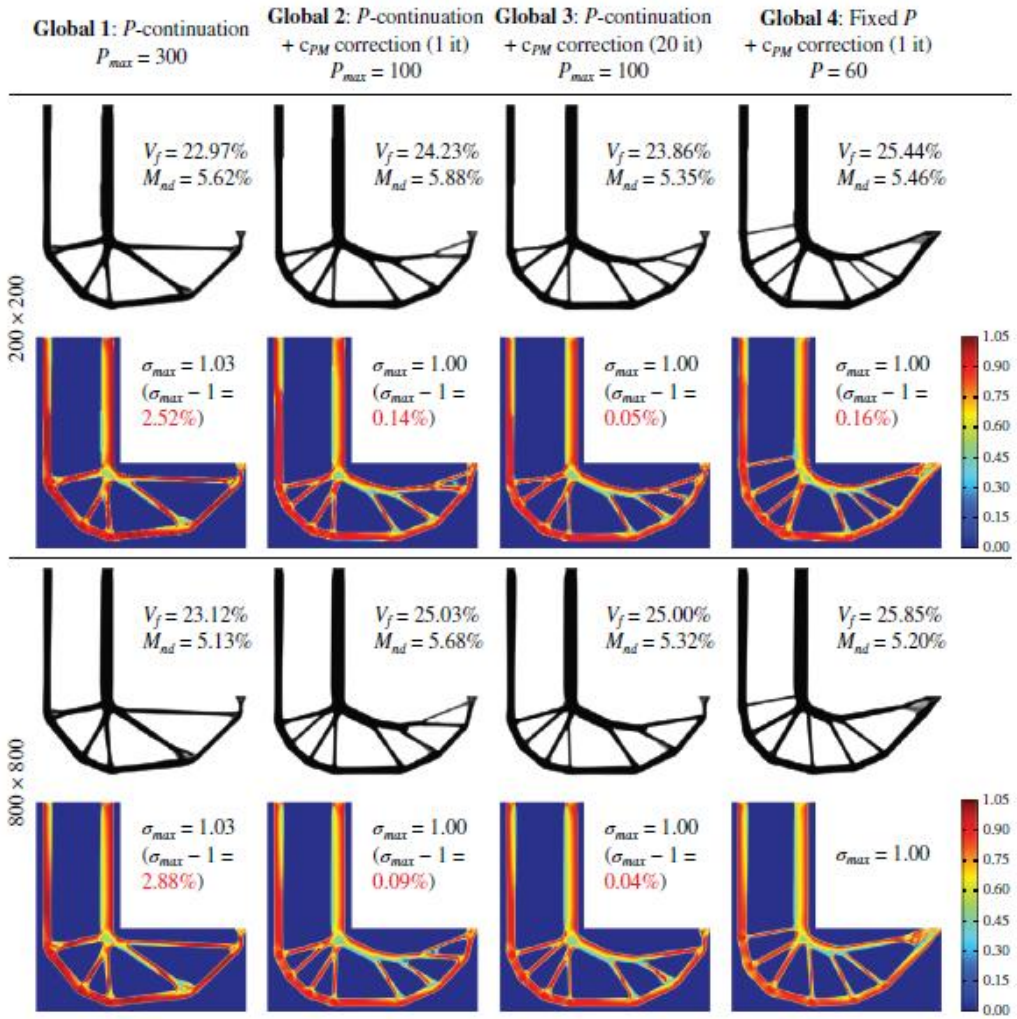


FIGURE 3 Optimized topologies and respective von Mises stresses obtained with the four global strategies for the most promising values of P_{max} and two mesh resolutions, at 2000 iterations

- The global strategy show that an optimal value of P_{max} exist and is different for each strategy.
  - A too high P_{max} may lead to convergence issues and sub-optimal solutions.
  - Increasing P_{max} too much lead to higher V_f.
- The adaptive normalization strategy generally seem to work well and lead to smaller constraint violation.

Stress constraint feasibility

Change in design variables

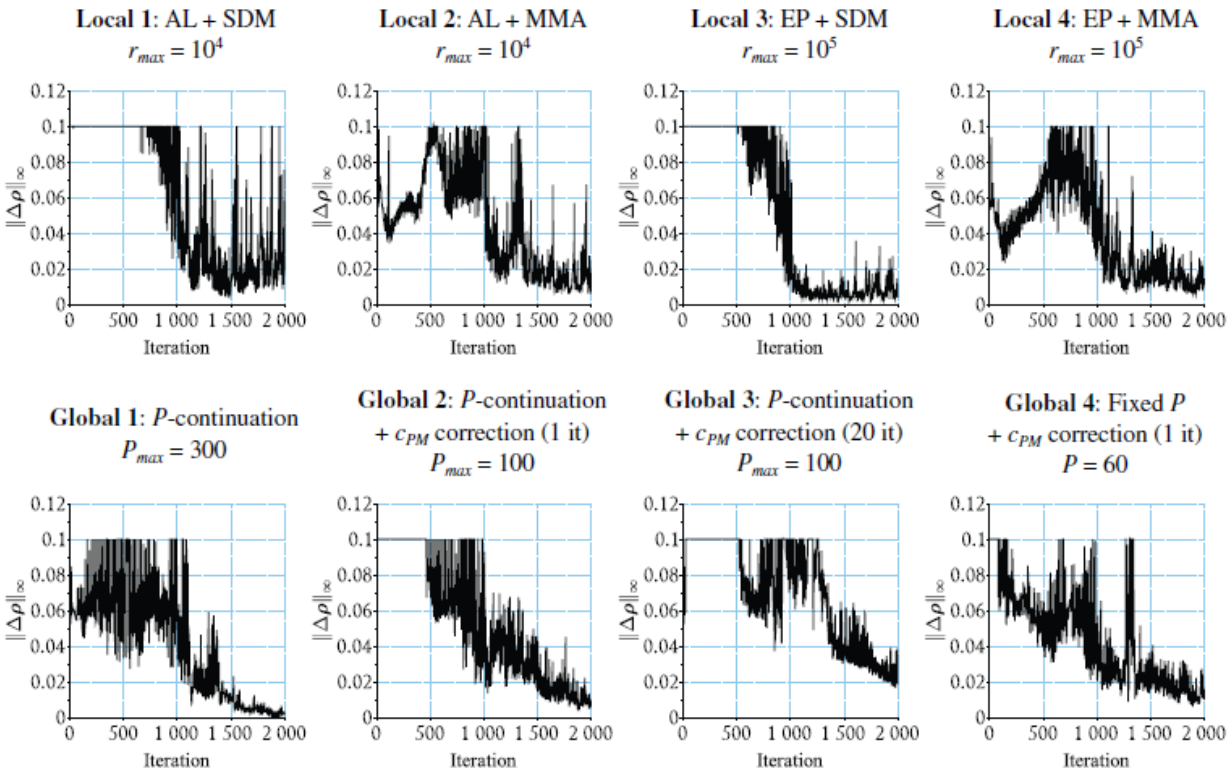


FIGURE 4 Maximum change iteration histories for all local and global solution strategies, for the most promising values of r_{max} and P_{max}, respectively, and 800 × 800 mesh size

Stabilization phase

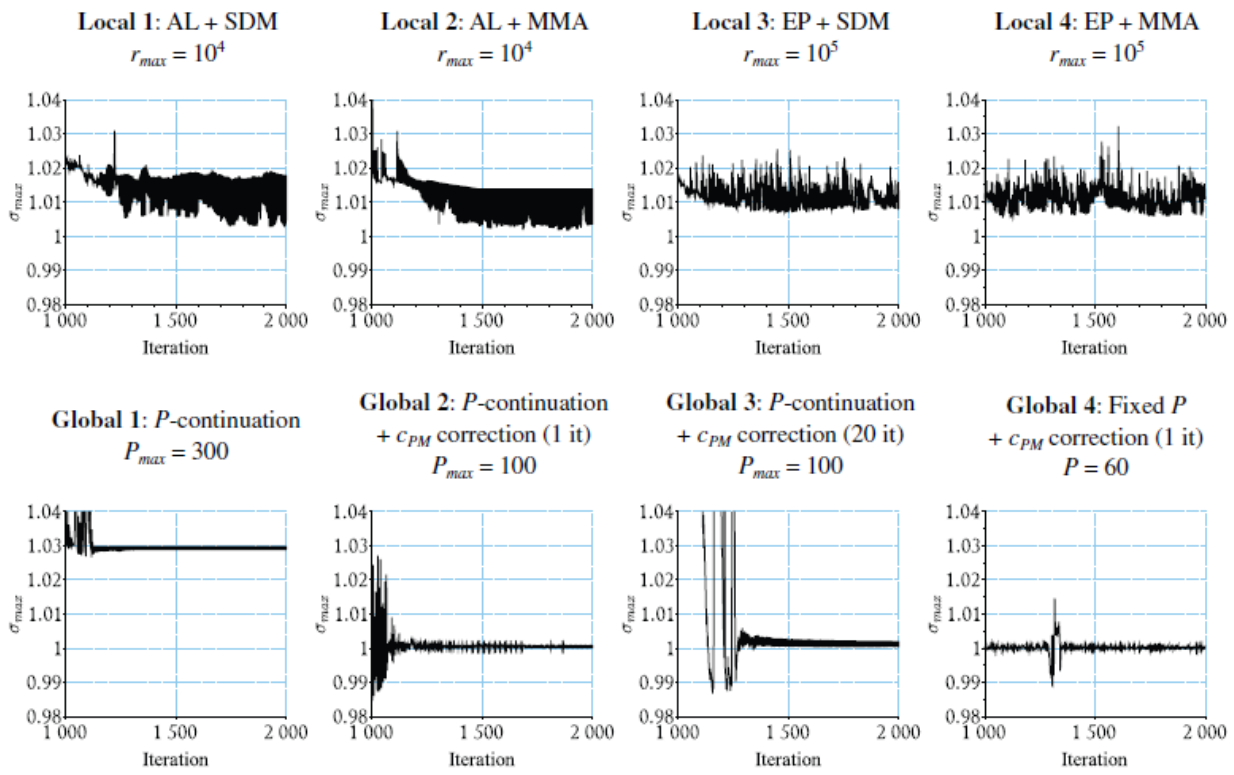


FIGURE 5 Maximum von Mises stresses for all local and global solution strategies, for the most promising values of r_{max} and P_{max}, respectively, and 800 × 800 mesh size

- Local strategy 3 (and 4) and global strategies 1-4 seems to converge most 'nicely' with $\Delta\rho \rightarrow 0$.
- Global strategies 2-4 satisfy the constraint quite accurately in the stabilization phase.
- A simple strategy to ensure the constraint is satisfied is to adjust the 'safety factor', α, to a value <1.

Recommended parameters

- Local 1 and 2: $\alpha = 0.98$, $r_{max}^1 = 10^4$ (used in phase 1), $r_{max}^2 = 10^5$(used in phase 2). During the stabilization phase, $\boldsymbol{\mu}$ and *r* are updated each 20 iterations only if σ_{max}>1.
- Local 3 and 4: $\alpha = 0.98$ and $r_{max} = 10^5$.
- Global 1: $\alpha = 0.97$ and $P_{max} = 300$.
- Global 2 and 3: $\alpha = 1.00$ and $P_{max} = 100$.
- Global 4. $\alpha = 1.00$ and $P = 60$ (fixed).

High resolution L shape – local strategies

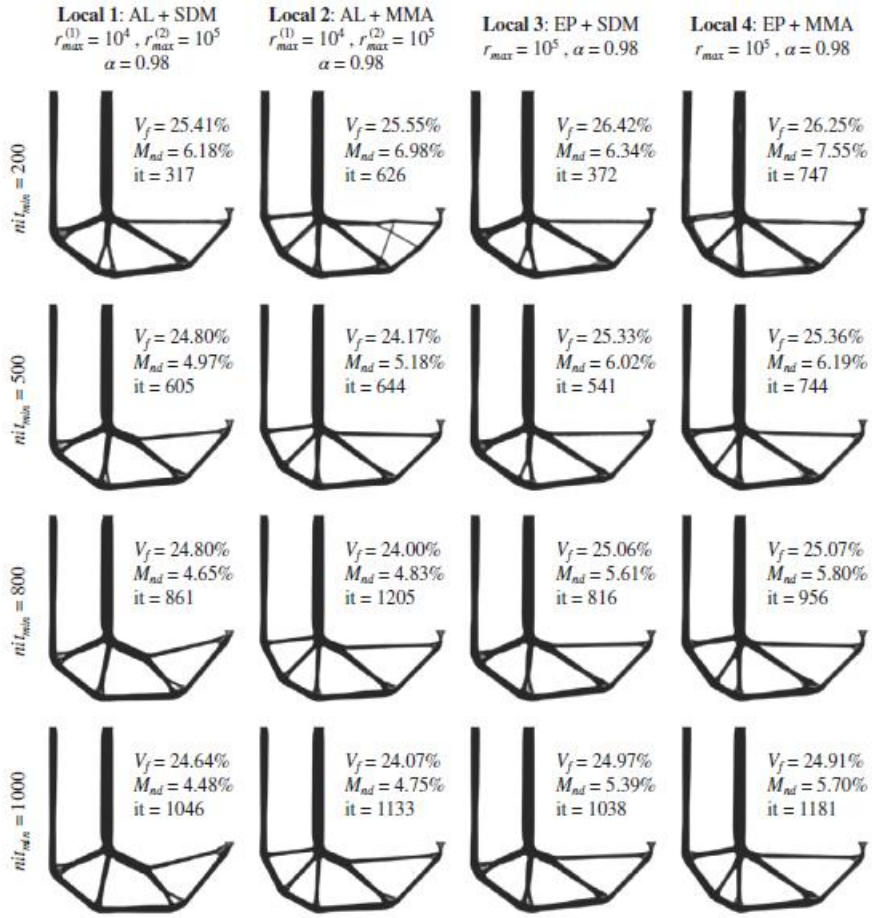


FIGURE 7 Optimized topologies obtained with the four local strategies using stabilization and feasibility stopping criteria, 800 × 800 mesh size

TABLE 3 Volume fractions, maximum stress constraint violations, number of iterations, and run times for the cases in Figures 7 and 8, obtained with the four local strategies

Solution strategy	nit_{min}	V_f (%)	$\sigma_{max} - 1$ (%)	Iterations	Run time (h)
Local 1: AL + SDM	200	25.41	−1.23	317	0.8
	500	24.80	−0.39	605	1.5
	800	24.80	−0.38	861	1.9
	1000	24.64	−0.39	1046	2.2
Local 2: AL + MMA	200	25.55	−1.02	626	2.0
	500	24.17	−0.45	644	1.6
	800	24.00	−0.28	1205	3.1
	1000	24.07	−0.57	1133	2.6
Local 3: EP + SDM	200	26.42	−1.37	372	1.1
	500	25.33	−0.54	541	1.3
	800	25.06	−0.20	816	1.8
	1000	24.97	−0.31	1038	2.3
Local 4: EP + MMA	200	26.25	−0.64	747	2.3
	500	25.36	−0.96	744	2.0
	800	25.07	−0.48	956	2.1
	1000	24.91	−1.14	1181	2.8

- Different number of iterations is used in phase 1 and phase 2 is completed if some convergence criteria are met.
- The local strategies show good results and are quite insensitive to the number of iterations in phase 1.
- Local 2 seems to give the smallest material usage, V_f.
- Local 1 seems to give a bit higher material usage but uses less iterations.

High resolution L shape – global strategies

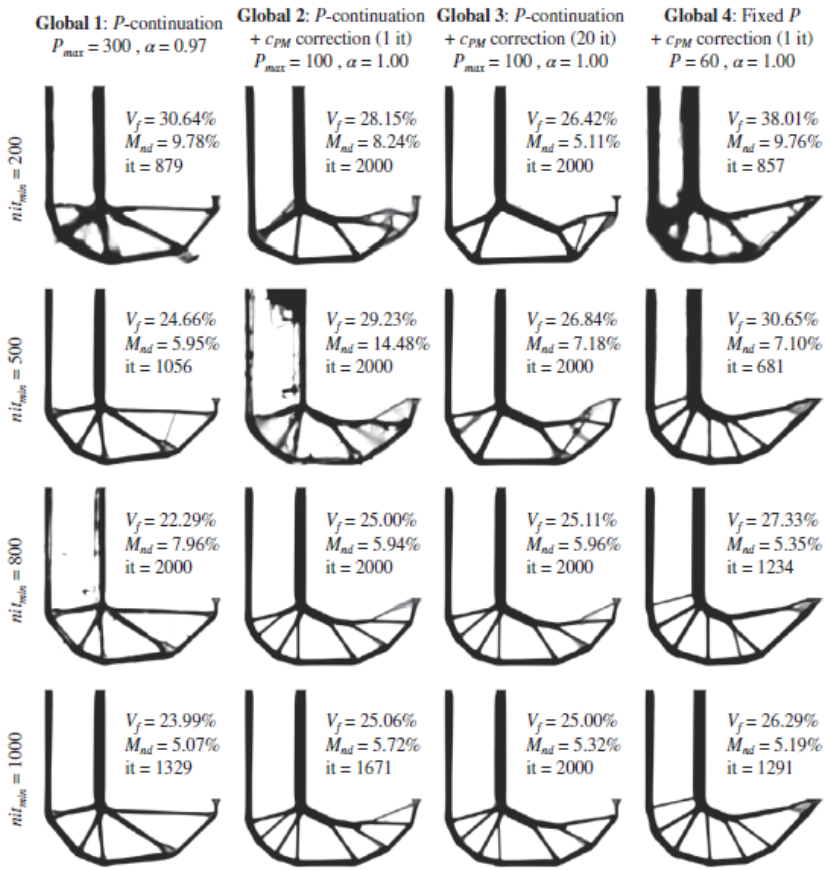


FIGURE 9 Optimized topologies obtained with the four global strategies using stabilization and feasibility stopping criteria, 800 × 800 mesh size

TABLE 4 Volume fractions, maximum stress constraint violations, number of iterations, and run times for the cases in Figures 9 and 10, obtained with the four global strategies

Solution strategy	nit_{min}	V_f (%)	$\sigma_{max} - 1$ (%)	Iterations	Run time (h)
Global 1: *P*-continuation	200	30.64	−0.03	879	2.8
	500	24.66	−0.05	1056	3.7
	800	22.29	63.19	2000	11.3
	1000	23.99	−0.15	1329	3.8
Global 2: *P*-continuation + c_{PM} (1 it)	200	28.15	0.10	2000	6.4
	500	29.23	51.99	2000	7.7
	800	25.00	0.04	2000	5.8
	1000	25.06	0.00	1671	4.7
Global 3: *P*-continuation + c_{PM} (20 it)	200	26.42	0.05	2000	6.6
	500	26.84	0.14	2000	6.4
	800	25.11	0.05	2000	6.0
	1000	25.00	0.04	2000	5.9
Global 4: Fixed *P* + c_{PM} (1 it)	200	38.01	−0.10	857	2.8
	500	30.65	−0.80	681	2.0
	800	27.33	−0.05	1234	3.6
	1000	26.29	−0.26	1291	3.5

- Different number of iterations is used in phase 1 and phase 2 is completed if some convergence criteria are met.
- The global strategies show some sensitivity to the number of iterations in phase 1 is observed.
- Some very large constraint violations are seen in some cases.
- The maximum number of iterations is reached in many cases.
- The lowest value of V_f of all strategies is Global 1 with 1000 iterations in phase 1.
- The local strategies generally require less iterations to converge than the global strategies, and the global strategies require a more careful continuation strategy of *P*.

Crack problem

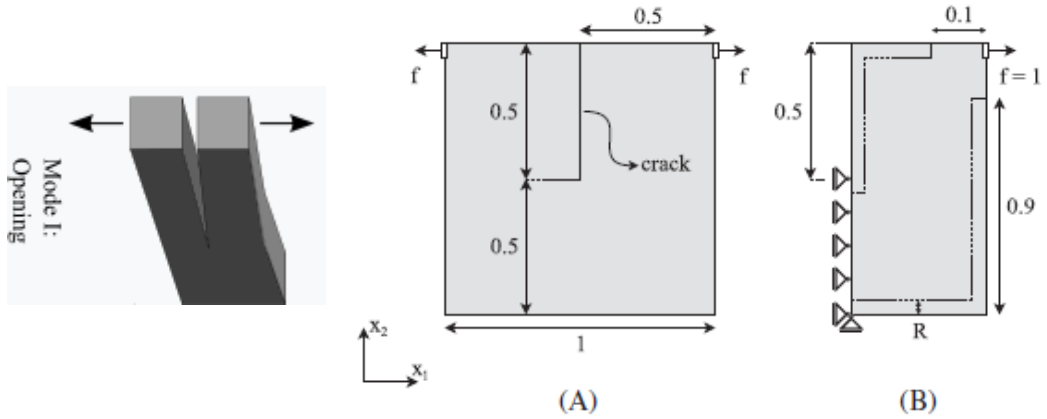


FIGURE 12 (A) Crack problem and (B) design domain employed in optimization (exploiting symmetry). The padding region is represented by dashed lines

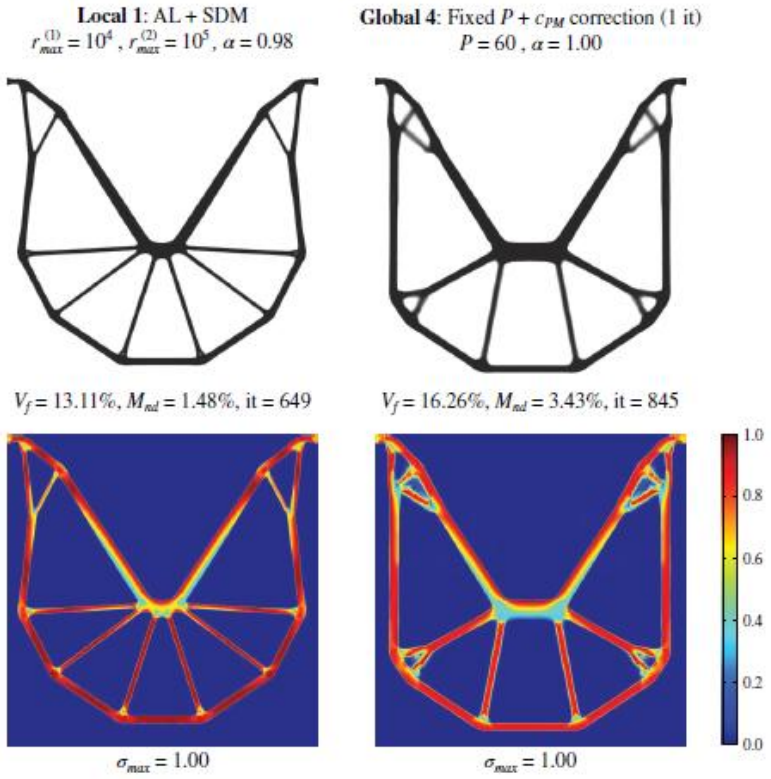


FIGURE 13 Crack problem. Optimized topologies and respective von Mises stress fields obtained with Local 1 (left) and Global 4 (right) strategies, for $nit_{min} = 500$. Half domain resolution: 1600×3200 ($N_e = 5,120,000$)

- Local 1 strategy reached an optimized topology in less iterations and with smaller volume fraction when compared with the Global 4 strategy result.
- Global 4 strategy show blurred structural members next to the region where the loads are applied.
- Local 1 strategy is result in a more well-defined geometry.
- The structural members of the Global 4 strategy have few points close to the yield stress value, a strong indicator that its structural volume could be further minimized while keeping the stress constraint satisfied.

Conclusions

"Based on results presented herein, we conclude that stress-constrained topology optimization should be addressed by local strategies. The AL-based strategies (Local 1 and 2) are very promising and should be preferable over global ones, since they provide better efficiency and performance for medium- and large-scale problems."

- Generally, both local and global strategies can be used to include stress constraints with reasonable results.
  - Use a safety factor if needed.
- The global *P*-mean aggregation method may be the easiest to implement in FEniCS.
- The local augmented Lagrangian or penalty methods should also be possible to implement.
  - A steepest descent implementation is available upon request ☺
  - You can also use MMA with a dummy constraint or volume fraction constraint.
- Tuning of the different methods can provide different results and final geometries.
  - This happens in many problems, not only stress constrained problems!
- Stress-constraints are quite difficult to handle, and care should be taken in the implementation and verification!
  - Always check your optimized design in an independent FEM calculation with a geometry derived from a CAD model and with a converged mesh!